Experiment 1

Title:- Time Series Data Import, Visualization and Descriptive Statistics

Aim:- To import a time series dataset into R, visualize the data using various plots, and calculate descriptive statistics for understanding the characteristics of the time series.

Procedure:

Step 1: Load the Dataset

Since **AirPassengers** is already available in R, simply load it.

```
# Load the dataset
data("AirPassengers")

# View dataset
AirPassengers
```

Step 2: Check Dataset Structure

```
# Convert to time series object
ts_data <- ts(AirPassengers, start = c(1949,1), frequency = 12)
```

Step 3: Check Dataset Structure

```
# Check structure
str(ts_data)

# Class
class(ts_data)

# Start Year
start(ts_data)

# End Year
end(ts_data)

# Frequency
frequency(ts_data)
```

Step 4: Display First and Last Observations

```
head(ts_data)
tail(ts_data)
```

Step 5: Statistics Operations

5.1 Statistics Summary

```
summary(AirPassengers)
```

5.2 Calculate Mean

```
mean(ts_data)
```

5.2 Calculate Median

```
median(ts_data)
```

5.3 Standard Deviation

```
sd(ts_data)
```

5.4 Find Minimum and Maximum

```
min(ts_data)
max(ts_data)
```

5.5 Find Range

```
range(ts_data)
```

5.6 Find Quantiles

```
quantile(ts_data)
```

Step 6: Check Missing Value

```
sum(is.na(ts_data))
```

Step 7: Draw Line Plot

```
plot(ts_data,
type="l",
col="red",
lwd=2,
main="Line Plot of AirPassengers")
```

Step 8: Draw Histogram

```
hist(ts_data,
    main="Histogram of AirPassengers",
    col="skyblue",
    border="black",
    xlab="Passengers")
```

Step 9: Box Plot

```
boxplot(AirPassengers,
      main="Boxplot of AirPassengers",
      col="lightgreen")
```

Step 10: Time Plot with grid

```
plot(AirPassengers,
    main="Time Series Plot",
    col="darkgreen",
    lwd=2)
grid()
```